Final Exam

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Question 1

Use the "Boston Housing" data set. Below are the data and variable descriptions:

Variable Information:

Number of Attributes: 13 continuous variables, and 1 binary-valued variable.

CRIM: per capita crime rate by town
ZN: proportion of residential land zoned for lots over 25,000 sq.ft.
INDU: proportion of non-retail business acres per town
CHAS: Charles River dummy variable (= 1 if tract bounds river; 0 otherwise)
NOX: nitric oxides concentration (parts per 10 million)
RM: average number of rooms per dwelling
AGE: proportion of owner-occupied units built prior to 1940
DIS: weighted distances to five Boston employment centres
RAD: index of accessibility to radial highways
TAX: full-value property-tax rate per $10,000
PTRATIO: pupil-teacher ratio by town
B: 1000(Bk - 0.63)^2 where Bk is the proportion of blacks by town
LSTAT: % lower status of the population
MEDV: Median value of owner-occupied homes in $1000's

(65 points)

a-) Use "set.seed(2546)" to create development sample (70% of the data) and hold-out sample (30% of the data). (5 points)

We first start examining the dataset and find out that there is no missing value. We notice that 'CHAS' is a categorical variable with 0 and 1, so there is no issue there, and 'RAD' looks like a categorical variable, but since it is described as an index of accessibility, we regard its values actually have numerical meanings.

```
housing <- getDataSet("Boston_Housing_Data.csv")
str(housing)
```

```
## 'data.frame':    506 obs. of  14 variables:
##  $ CRIM   : num  0.00632 0.02731 0.02729 0.03237 0.06905 ...
##  $ ZN     : num  18 0 0 0 0 0 12.5 12.5 12.5 12.5 ...
##  $ INDUS  : num  2.31 7.07 7.07 2.18 2.18 2.18 7.87 7.87 7.87 7.87 ...
##  $ CHAS   : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ NOX    : num  0.538 0.469 0.469 0.458 0.458 0.458 0.524 0.524 0.524 0.524 ...
##  $ RM     : num  6.58 6.42 7.18 7 7.15 ...
##  $ AGE    : num  65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
##  $ DIS    : num  4.09 4.97 4.97 6.06 6.06 ...
##  $ RAD    : int  1 2 2 3 3 3 5 5 5 5 ...
```

```
##  $ TAX    : int  296 242 242 222 222 222 311 311 311 311 ...
##  $ PTRATIO: num  15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 15.2 ...
##  $ B      : num  397 397 393 395 397 ...
##  $ LSTAT  : num  4.98 9.14 4.03 2.94 5.33 ...
##  $ MEDV   : num  24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...
```

```
head(housing)
```

```
##      CRIM ZN INDUS CHAS   NOX    RM  AGE    DIS RAD TAX PTRATIO      B LSTAT
## 1 0.00632 18  2.31    0 0.538 6.575 65.2 4.0900   1 296    15.3 396.90  4.98
## 2 0.02731  0  7.07    0 0.469 6.421 78.9 4.9671   2 242    17.8 396.90  9.14
## 3 0.02729  0  7.07    0 0.469 7.185 61.1 4.9671   2 242    17.8 392.83  4.03
## 4 0.03237  0  2.18    0 0.458 6.998 45.8 6.0622   3 222    18.7 394.63  2.94
## 5 0.06905  0  2.18    0 0.458 7.147 54.2 6.0622   3 222    18.7 396.90  5.33
## 6 0.02985  0  2.18    0 0.458 6.430 58.7 6.0622   3 222    18.7 394.12  5.21
##   MEDV
## 1 24.0
## 2 21.6
## 3 34.7
## 4 33.4
## 5 36.2
## 6 28.7
```

```
apply(housing,2,function(x) sum(is.na(x)))
```

```
##    CRIM      ZN   INDUS    CHAS     NOX      RM     AGE     DIS     RAD     TAX
##       0       0       0       0       0       0       0       0       0       0
## PTRATIO       B   LSTAT    MEDV
##       0       0       0       0
```

```
table(housing$CHAS)
```

```
##
##   0   1
## 471  35
```

```
table(housing$RAD)
```

```
##
##   1   2   3   4   5   6   7   8  24
##  20  23  43 109 113  26  17  23 132
```

```
set.seed(2546)
train_index <- createDataPartition(y = housing$CRIM, p = 0.7, list = FALSE)
train_set <- housing[train_index,]
test_set <- housing[-train_index,]
```

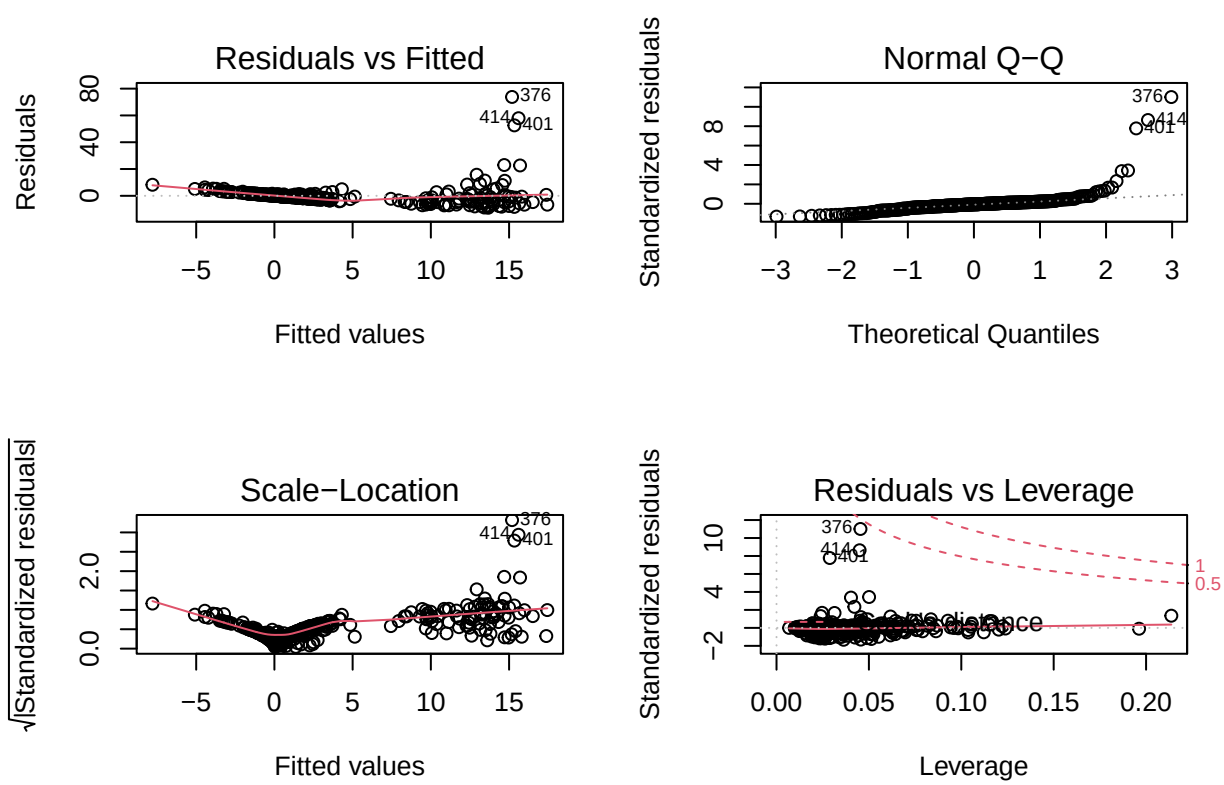
b-) Built a regression model to predict Crime Rate (CRIM) on the development sample by using all the variables. Do we need to transform CRIM? If so, transform the CRIM and redo the regression model. (5 points)

```
f1 <- lm(CRIM ~ ., data=train_set)
summary(f1)
```

```
##
## Call:
## lm(formula = CRIM ~ ., data = train_set)
##
```

```
## Residuals:
##    Min     1Q Median     3Q    Max
## -8.911 -2.125 -0.379  0.847 73.786
##
## Coefficients:
##               Estimate Std. Error t value    Pr(>|t|)
## (Intercept) 10.6928685  9.1790130   1.165    0.244861
## ZN           0.0441563  0.0250528   1.763    0.078872 .
## INDUS       -0.0318672  0.1234282  -0.258    0.796420
## CHAS        -0.8210472  1.4346484  -0.572    0.567496
## NOX         -9.3070107  6.8209588  -1.364    0.173316
## RM           1.2586761  0.8577169   1.467    0.143167
## AGE          0.0009037  0.0223212   0.040    0.967729
## DIS         -0.9910845  0.3641332  -2.722    0.006826 **
## RAD          0.5637476  0.1128888   4.994 0.000000945 ***
## TAX         -0.0036618  0.0071177  -0.514    0.607262
## PTRATIO     -0.2419866  0.2235729  -1.082    0.279855
## B           -0.0036707  0.0046877  -0.783    0.434135
## LSTAT        0.1001321  0.0945820   1.059    0.290493
## MEDV        -0.2500008  0.0739924  -3.379    0.000812 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.863 on 342 degrees of freedom
## Multiple R-squared:  0.4106, Adjusted R-squared:  0.3882
## F-statistic: 18.33 on 13 and 342 DF,  p-value: < 2.2e-16
```

```r
par(mfrow=c(2,2))
plot(f1)
```

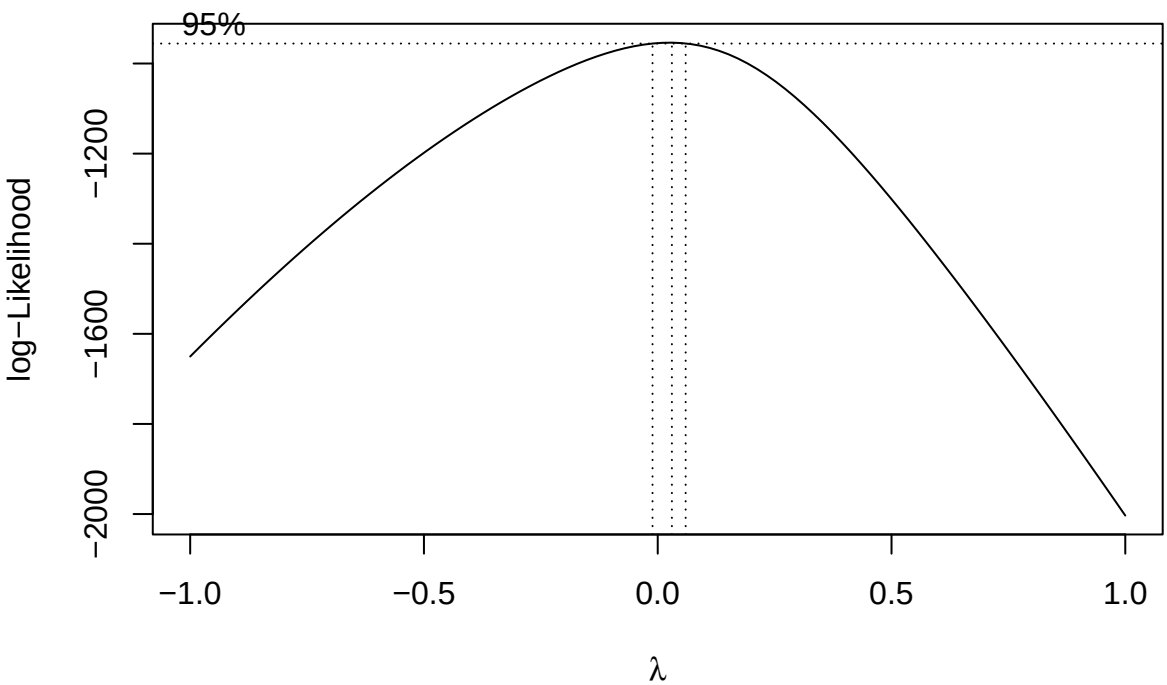


```r
shapiro.test(f1$residuals)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  f1$residuals
## W = 0.45633, p-value < 2.2e-16
```

The regression model gives us R^2 of 41.06%, and 'DIS', 'RAD', 'MEDV' are indicated as significant. The QQ plot and the Shapiro-Wilk test show that the error terms are not normally distributed. The Breuch-Pagan test indicates that the errors do not have constant variance. Therefore, we need the transformation on the response variable, and the Box-Cox procedure below suggests the log transformation. After the log transformation, the QQ plot is approximately normally distributed, the residual vs fitted values plot also improved a lot and looks more constant.

```r
bc <-boxcox(f1, lambda=seq(-1,1,0.025))
```




```r
lambda <- bc$x[which.max(bc$y)]
lambda
```

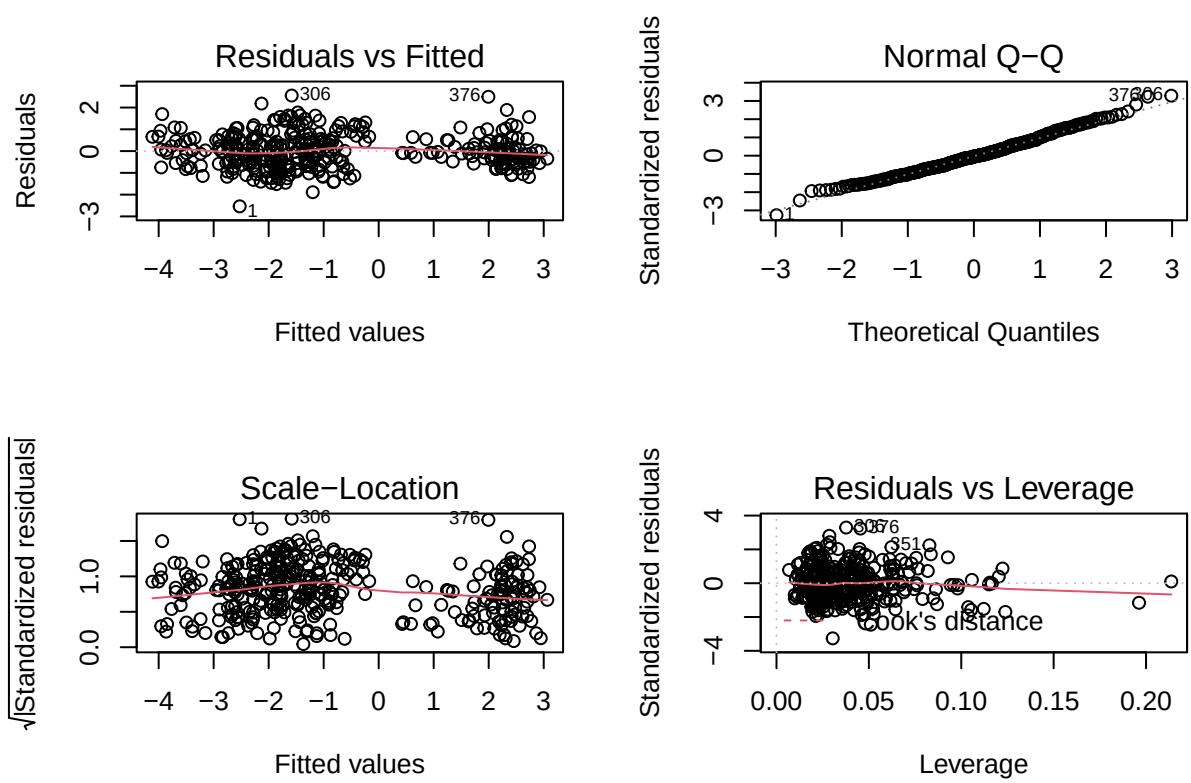
```
## [1] 0.03030303
```

```r
f1_log <- lm(log(CRIM) ~ ., data=train_set)
summary(f1_log)
```

```
##
## Call:
## lm(formula = log(CRIM) ~ ., data = train_set)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.53729 -0.54096 -0.03869  0.50050  2.55028
##
## Coefficients:
```

```
##               Estimate Std. Error t value  Pr(>|t|)
## (Intercept) -2.9368439  1.0565116  -2.780   0.00574 **
## ZN          -0.0092741  0.0028836  -3.216   0.00142 **
## INDUS        0.0442830  0.0142067   3.117   0.00198 **
## CHAS        -0.2003513  0.1651292  -1.213   0.22585
## NOX          3.3454604  0.7850977   4.261 0.0000263 ***
## RM          -0.0732378  0.0987239  -0.742   0.45869
## AGE          0.0080926  0.0025692   3.150   0.00178 **
## DIS         -0.0277614  0.0419120  -0.662   0.50818
## RAD          0.1570207  0.0129936  12.084   < 2e-16 ***
## TAX         -0.0016088  0.0008193  -1.964   0.05037 .
## PTRATIO     -0.0472987  0.0257334  -1.838   0.06693 .
## B           -0.0013641  0.0005396  -2.528   0.01192 *
## LSTAT        0.0232063  0.0108865   2.132   0.03375 *
## MEDV         0.0087924  0.0085166   1.032   0.30262
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7899 on 342 degrees of freedom
## Multiple R-squared:  0.8674, Adjusted R-squared:  0.8623
## F-statistic:   172 on 13 and 342 DF,  p-value: < 2.2e-16
```

```r
par(mfrow=c(2,2))
plot(f1_log)
```



c-) Use stepwise (both ways) model selection to select the best model for predicting the transformed CRIM on the development sample. **Ensure that all variables are significant, use $\alpha = 0.05$. Justify your choice of model. Check all the regression model assumptions visually from the graphs. Perform a statistical test for auto correlation.There could be potential outliers in the model. Build a robust regression model by using the same variables in your final model and compare your**

regression model and outputs with the best model. Do you see a significant impact of outliers? Please discuss your findings (15 points)

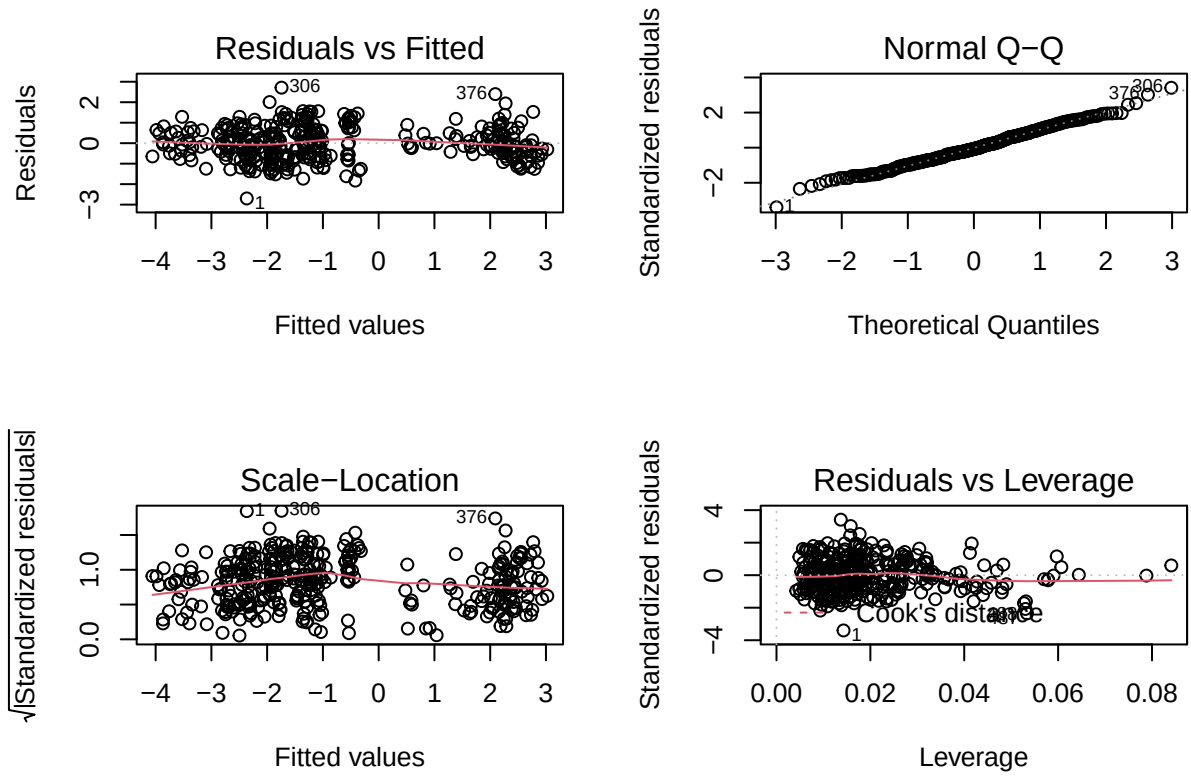
```
ols_step_both_p(f1_log, prem=0.05)
```

```
##
##                                Stepwise Selection Summary
## -----------------------------------------------------------------------------------------
##                       Added/                    Adj.
## Step    Variable    Removed     R-Square    R-Square      C(p)          AIC        RMSE
## -----------------------------------------------------------------------------------------
##    1       RAD       addition       0.717       0.716    377.5970    1103.8051    1.1340
##    2       NOX       addition       0.829       0.828     91.6170     927.0752    0.8835
##    3        ZN       addition       0.846       0.845     49.5700     891.6702    0.8395
##    4       AGE       addition       0.855       0.853     28.4880     872.3769    0.8159
##    5        B        addition       0.858       0.856     21.1850     865.4214    0.8068
##    6      INDUS      addition       0.861       0.859     15.7230     860.0722    0.7997
##    7       TAX       addition       0.863       0.860     13.9490     858.2964    0.7966
##    8       TAX       removal        0.861       0.859     15.7230     860.0722    0.7997
##    9      LSTAT      addition       0.863       0.860     14.1980     858.5464    0.7969
##   10      LSTAT      removal        0.861       0.859     15.7230     860.0722    0.7997
##   11     PTRATIO     addition       0.863       0.860     14.5060     858.8565    0.7972
##   12     PTRATIO     removal        0.861       0.859     15.7230     860.0722    0.7997
## -----------------------------------------------------------------------------------------
```

```
f1_best <- lm(log(CRIM) ~ RAD+NOX+ZN+AGE+B+INDUS, data=train_set)
summary(f1_best)
```

```
##
## Call:
## lm(formula = log(CRIM) ~ RAD + NOX + ZN + AGE + B + INDUS, data = train_set)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.69857 -0.56237 -0.05655  0.54099  2.71099
##
## Coefficients:
##               Estimate Std. Error t value     Pr(>|t|)
## (Intercept) -4.5652092  0.3870424 -11.795      < 2e-16 ***
## RAD          0.1312990  0.0068775  19.091      < 2e-16 ***
## NOX          3.9679032  0.7007052   5.663 0.0000000312 ***
## ZN          -0.0101356  0.0023809  -4.257 0.0000266535 ***
## AGE          0.0098283  0.0022510   4.366 0.0000166782 ***
## B           -0.0014931  0.0005138  -2.906      0.00390 **
## INDUS        0.0294244  0.0109057   2.698      0.00731 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7997 on 349 degrees of freedom
## Multiple R-squared:  0.8613, Adjusted R-squared:  0.8589
## F-statistic: 361.1 on 6 and 349 DF,  p-value: < 2.2e-16
```

```
par(mfrow=c(2,2))
plot(f1_best)
```




```r
shapiro.test(f1_best$residuals)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  f1_best$residuals
## W = 0.99435, p-value = 0.2117
```

```r
# Robust regression
rr.huber <- rlm(log(CRIM) ~ RAD+NOX+ZN+AGE+B+INDUS, data = train_set, psi = psi.huber)
summary(rr.huber)
```

```
##
## Call: rlm(formula = log(CRIM) ~ RAD + NOX + ZN + AGE + B + INDUS, data = train_set,
##     psi = psi.huber)
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.68935 -0.55450 -0.04777  0.55536  2.72002
##
## Coefficients:
##             Value    Std. Error t value
## (Intercept)  -4.6015   0.4107   -11.2045
## RAD           0.1292   0.0073    17.7060
## NOX           3.9314   0.7435     5.2878
## ZN           -0.0097   0.0025    -3.8255
## AGE           0.0099   0.0024     4.1431
## B            -0.0014   0.0005    -2.6016
## INDUS         0.0321   0.0116     2.7760
##
## Residual standard error: 0.8245 on 349 degrees of freedom
```

```r
# Check autocorrelation
dwtest(f1_best)
```

```
##
##  Durbin-Watson test
##
## data:  f1_best
## DW = 0.73035, p-value < 2.2e-16
## alternative hypothesis: true autocorrelation is greater than 0
```

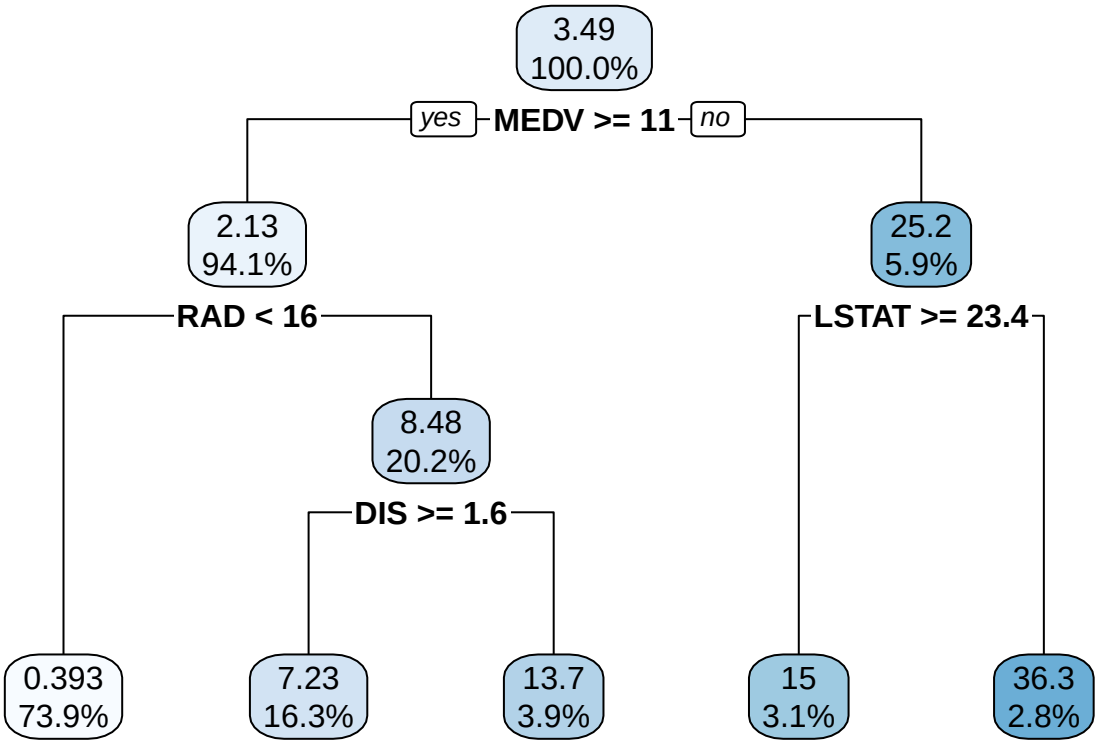
```r
f1_auto <- cochrane.orcutt(f1_best)
dwtest(f1_auto)
```

```
##
##  Durbin-Watson test
##
## data:  f1_auto
## DW = 2.2754, p-value = 0.9947
## alternative hypothesis: true autocorrelation is greater than 0
```

The stepwise both way model selects 'RAD','NOX','ZN','AGE','B', and 'INDUS' as predictors, and the summary of the model shows that all variables are significant. The residual vs fitted values plot shows constant variances of error terms, and the QQ plot shows the error are approximately normally distributed as well as the Shapiro-Wilk test. The coefficients of the robust regression model are quite close to our best model, which indicates that the outliers do not have significant impact. But, the Durbin-Watson test shows that there is autocorrelation, so we used the Cochrane-Orcutt method to remediate this issue, and finally we removed the autocorrelation as shown above.

d-) Use regression Tree to predict the CRIM on the development sample and compare the performance of tree against the regression model built in part c on development sample. (10 points)

```r
f1tree <- rpart(CRIM ~ ., data=train_set)
rpart.plot(f1tree, digits=3)
```




```
resampleMetrics(predict(f1tree, train_set), train_set$CRIM) # Regression Tree
```

```
##             R2     RMSE      MAE
## [1,] 0.6224104 5.383828 1.861058
```

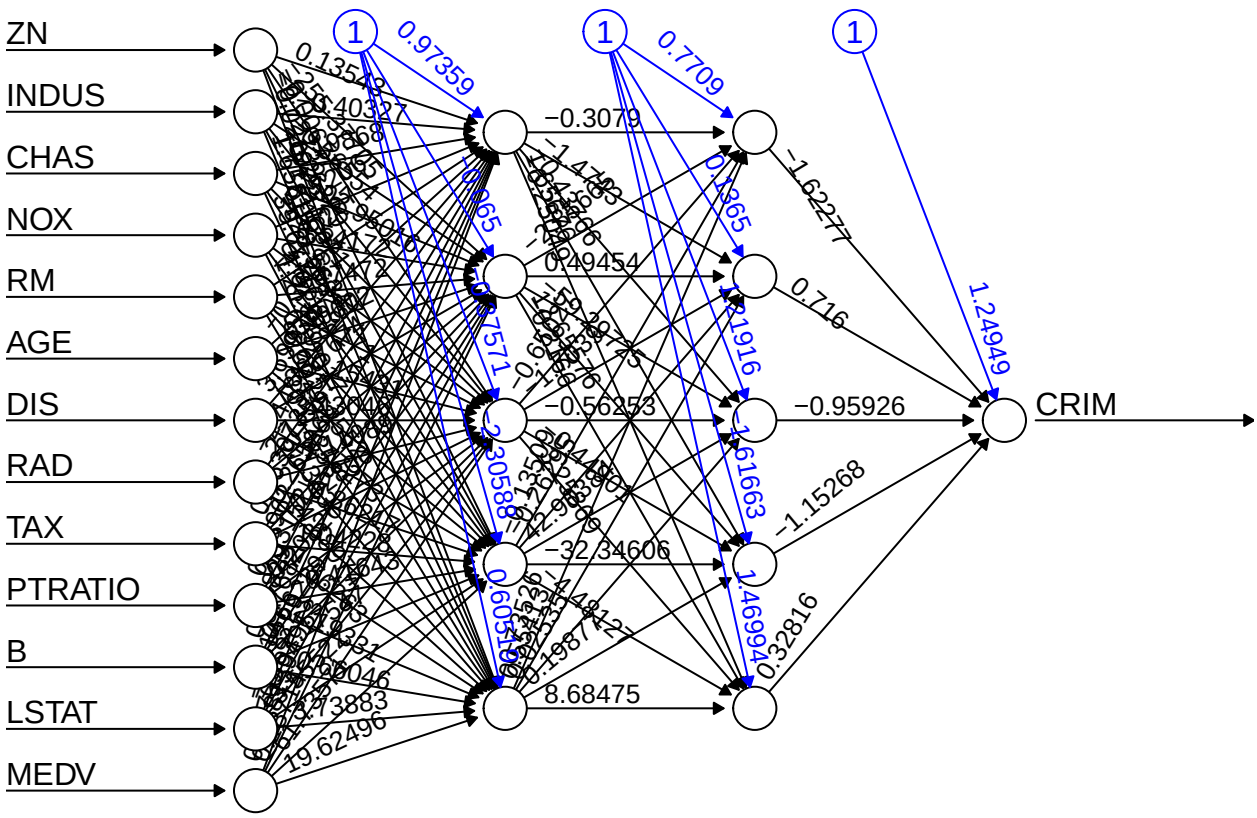
```
resampleMetrics(exp(predict(f1_best, train_set)), train_set$CRIM) # Stepwise
```

```
##             R2     RMSE      MAE
## [1,] 0.3413981 7.110378 1.999816
```

The regression tree has a higher R^2 and lower RMSE/MAE, so for now, the regression tree seems to be a better model.

e-) Use Neuron Network (NN) with two hidden layers with 5 neurons (hidden=c(5,5)) to predict the CRIM on the development sample and compare the performance of NN the regression model built in part c on development sample. (10 points)

```
normalize <- function(x) {return((x - min(x)) / (max(x) - min(x)))}
housing_norm <- as.data.frame(lapply(housing, normalize))
train_norm <- housing_norm[train_index,]
test_norm <- housing_norm[-train_index,]
NN <- neuralnet(CRIM ~ ., hidden=c(5,5), train_norm)
plot(NN, rep="best")
```




```
NN_result <- compute(NN, train_norm[,-c(1)])
denormalize <- function(x) {return((x * (max(housing$CRIM) - min(housing$CRIM))) + min(housing$CRIM))}
pred_NN <- denormalize(NN_result$net.result)

resampleMetrics(pred_NN, train_set$CRIM) # Neural Network
```

```
##             R2     RMSE       MAE
## [1,] 0.9565055 1.827251 0.8576451
```

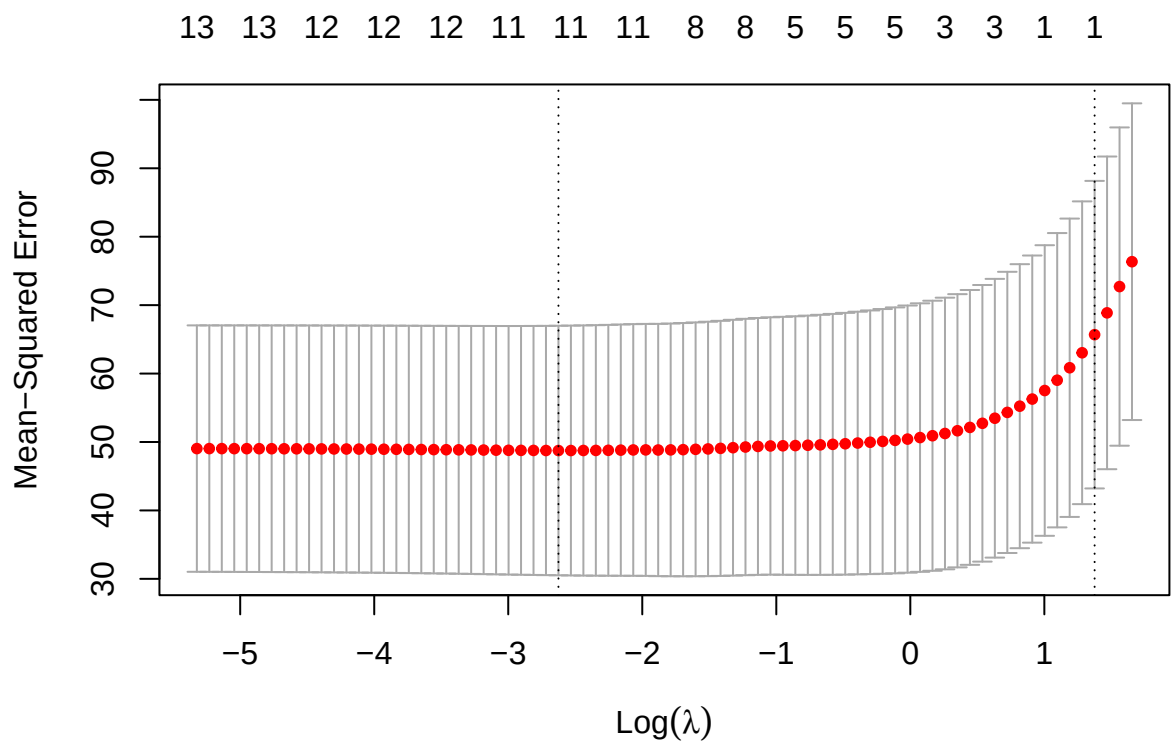
```
resampleMetrics(exp(predict(f1_best, train_set)), train_set$CRIM) # Stepwise
```

```
##             R2     RMSE      MAE
## [1,] 0.3413981 7.110378 1.999816
```

The neural network significantly outperforms with very high R^2 and lower RMSE/MAE.

f-) Use lasso regression to predict the CRIM on the development sample and compare the performance of lasso against the regression model built in part c on development sample. (10 points)

```
x <- model.matrix(CRIM~., train_set)[,-c(1)]
y <- train_set$CRIM
LassoMod <- glmnet(x, y, alpha=1, nlambda=100,lambda.min.ratio=0.0001)
CvLassoMod <- cv.glmnet(x, y, alpha=1, nlambda=100,lambda.min.ratio=0.0001)
plot(CvLassoMod)
```




```
best.lambda.lasso <- CvLassoMod$lambda.min
pred.lasso <- coef(CvLassoMod, s = "lambda.min")
pred.lasso
```

```
## 14 x 1 sparse Matrix of class "dgCMatrix"
##                      s1
## (Intercept)  5.34334781
## ZN           0.03356675
## INDUS       -0.03542166
## CHAS        -0.58403458
## NOX         -4.40977675
## RM           0.97470444
## AGE          .
## DIS         -0.71661877
## RAD          0.48654510
## TAX          .
## PTRATIO     -0.13762521
## B           -0.00365359
## LSTAT        0.08674039
## MEDV        -0.20355271
```

```
yhat.lasso <- predict(LassoMod, s = best.lambda.lasso, newx = x)

resampleMetrics(yhat.lasso, train_set$CRIM) # Lasso
```

```
##             R2     RMSE     MAE
## [1,] 0.4076089 6.743502 2.67457
```

```
resampleMetrics(exp(predict(f1_best, train_set)), train_set$CRIM) # Stepwise
```

```
##             R2     RMSE      MAE
## [1,] 0.3413981 7.110378 1.999816
```

The Lasso regression has a higher R^2 and lower RMSE. MAE is bigger, but we would still choose the lasso regression over the one from part(c).

g-) Evaluate the performances of models built in part c (use the model selected based on the stepwise model procedure) through part f on hold-out sample. Compare the SSEs, R^2 and select the best model. (10 points)

```
SSE <- function(actual, predicted) {sum((actual - predicted)^2)}

# Stepwise from part(c)
resampleMetrics(exp(predict(f1_best, test_set)), test_set$CRIM)
```

```
##             R2    RMSE      MAE
## [1,] 0.4847689 5.86155 2.064151
```

```
SSE(exp(predict(f1_best, test_set)), test_set$CRIM)
```

```
## [1] 5153.665
```

```
# Regression Tree
resampleMetrics(predict(f1tree, test_set), test_set$CRIM)
```

```
##             R2    RMSE      MAE
## [1,] 0.4859707 5.85471 2.165238
```

```
SSE(predict(f1tree, test_set), test_set$CRIM)
```

```
## [1] 5141.644
```

```
# Neural Network
set.seed(2546)
NN_result <- compute(NN, test_norm[,-c(1)])
pred_NN <- denormalize(NN_result$net.result)
resampleMetrics(pred_NN, test_set$CRIM)
```

```
##             R2     RMSE      MAE
## [1,] 0.6894515 4.550679 1.799619
```

```
SSE(pred_NN, test_set$CRIM)
```

```
## [1] 3106.301
```

```
# Lasso
x2 <- model.matrix(CRIM~., test_set)[,-c(1)]
yhat.lasso <- predict(LassoMod, s = best.lambda.lasso, newx = x2)
resampleMetrics(yhat.lasso, test_set$CRIM)
```

```
##             R2    RMSE      MAE
## [1,] 0.5448916 5.50895 2.784585
```

```
SSE(yhat.lasso, test_set$CRIM)
```

```
## [1] 4552.28
```

Based on the SSE and R^2, we choose the neural network since it has the highest R^2 and lowest SSE.

Question 2

Use biopsy data sets in the MASS package. Copy and paste and following code "library(MASS);data(biopsy,package ="MASS")".

the first variable is the ID variable and you need to drop it from the data

There are 699 cases of potentially cancerous tumors of which 241 are actually malignant. The response variable class shows if tumor is malignant or not (0 if malignant, 1 if benign). (20 points, 5 points each)

a-) Fit a regression model to predict Class and comment on your model. Can we drop $V2$ and $V9$?

We dropped the first column 'ID', and since 'V6' has missing values, we dropped them by rows.

```
data(biopsy)
biopsy <- biopsy[,-1]
str(biopsy)
```

```
## 'data.frame':    699 obs. of  10 variables:
##  $ V1   : int  5 5 3 6 4 8 1 2 2 4 ...
##  $ V2   : int  1 4 1 8 1 10 1 1 1 2 ...
##  $ V3   : int  1 4 1 8 1 10 1 2 1 1 ...
##  $ V4   : int  1 5 1 1 3 8 1 1 1 1 ...
##  $ V5   : int  2 7 2 3 2 7 2 2 2 2 ...
##  $ V6   : int  1 10 2 4 1 10 10 1 1 1 ...
##  $ V7   : int  3 3 3 3 3 9 3 3 1 2 ...
##  $ V8   : int  1 2 1 7 1 7 1 1 1 1 ...
##  $ V9   : int  1 1 1 1 1 1 1 1 5 1 ...
##  $ class: Factor w/ 2 levels "benign","malignant": 1 1 1 1 1 2 1 1 1 1 ...
```

```
head(biopsy)
```

```
##   V1 V2 V3 V4 V5 V6 V7 V8 V9     class
## 1  5  1  1  1  2  1  3  1  1    benign
## 2  5  4  4  5  7 10  3  2  1    benign
## 3  3  1  1  1  2  2  3  1  1    benign
## 4  6  8  8  1  3  4  3  7  1    benign
## 5  4  1  1  3  2  1  3  1  1    benign
## 6  8 10 10  8  7 10  9  7  1 malignant
```

```
apply(biopsy,2,function(x) sum(is.na(x)))
```

```
##    V1    V2    V3    V4    V5    V6    V7    V8    V9 class
##     0     0     0     0     0    16     0     0     0     0
```

```
biopsy2 <- na.omit(biopsy)
biopsy2[,10] <- ifelse(biopsy2[,10]=="benign",1,0)

f2 <- glm(class ~ ., family=binomial, data=biopsy2)
summary(f2)
```

```
##
## Call:
## glm(formula = class ~ ., family = binomial, data = biopsy2)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -2.4698  -0.0222   0.0619   0.1153   3.4841
##
## Coefficients:
##             Estimate Std. Error z value  Pr(>|z|)
## (Intercept) 10.10394    1.17488   8.600   < 2e-16 ***
## V1          -0.53501    0.14202  -3.767  0.000165 ***
## V2           0.00628    0.20908   0.030  0.976039
## V3          -0.32271    0.23060  -1.399  0.161688
```

```
## V4          -0.33064    0.12345  -2.678  0.007400 **
## V5          -0.09663    0.15659  -0.617  0.537159
## V6          -0.38303    0.09384  -4.082 0.0000447 ***
## V7          -0.44719    0.17138  -2.609  0.009073 **
## V8          -0.21303    0.11287  -1.887  0.059115 .
## V9          -0.53484    0.32877  -1.627  0.103788
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 884.35  on 682  degrees of freedom
## Residual deviance: 102.89  on 673  degrees of freedom
## AIC: 122.89
##
## Number of Fisher Scoring iterations: 8
```

```r
f2r <- glm(class ~ V1+V3+V4+V5+V6+V7+V8, family=binomial, data=biopsy2)
anova(f2r, f2, test="Chi")
```

```
## Analysis of Deviance Table
##
## Model 1: class ~ V1 + V3 + V4 + V5 + V6 + V7 + V8
## Model 2: class ~ V1 + V2 + V3 + V4 + V5 + V6 + V7 + V8 + V9
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1       675     106.66
## 2       673     102.89  2   3.7706   0.1518
```

Since the response variable is categorical and binary, we fit the logistic regression model. 'V1','V4','V6', and 'V7' are indicated as significant. The ANOVA table above shows the p-value of 0.1518, so we accept the null hypothesis that 'V2' and 'V9' can be dropped.

b-) Perform variable selection (use step function with using stepwise both direction method). (Hint: Use the step function.) **Ensure that all variables are significant, use $\alpha = 0.05$.**

```r
f2_step <- step(f2, direction="both", trace=0)
summary(f2_step)
```

```
##
## Call:
## glm(formula = class ~ V1 + V3 + V4 + V6 + V7 + V8 + V9, family = binomial,
##     data = biopsy2)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -2.4115  -0.0219   0.0627   0.1149   3.5235
##
## Coefficients:
##             Estimate Std. Error z value  Pr(>|z|)
## (Intercept)  9.98278    1.12610   8.865   < 2e-16 ***
## V1          -0.53400    0.14079  -3.793  0.000149 ***
## V3          -0.34529    0.17164  -2.012  0.044255 *
## V4          -0.34249    0.11922  -2.873  0.004068 **
## V6          -0.38830    0.09356  -4.150 0.0000332 ***
## V7          -0.46194    0.16820  -2.746  0.006025 **
## V8          -0.22606    0.11097  -2.037  0.041644 *
```

```
## V9          -0.53119    0.32446  -1.637  0.101598
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 884.35  on 682  degrees of freedom
## Residual deviance: 103.27  on 675  degrees of freedom
## AIC: 119.27
##
## Number of Fisher Scoring iterations: 8
```

```
drop1(f2_step, test="Chi")
```

```
## Single term deletions
##
## Model:
## class ~ V1 + V3 + V4 + V6 + V7 + V8 + V9
##        Df Deviance    AIC     LRT    Pr(>Chi)
## <none>      103.27 119.27
## V1      1   121.55 135.55 18.2850 0.000019019 ***
## V3      1   107.90 121.90  4.6353    0.031320 *
## V4      1   112.17 126.17  8.9070    0.002841 **
## V6      1   123.15 137.15 19.8871 0.000008215 ***
## V7      1   111.69 125.69  8.4219    0.003707 **
## V8      1   107.72 121.72  4.4551    0.034798 *
## V9      1   107.14 121.14  3.8770    0.048953 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The stepwise both direction method dropped 'V2' and 'V5', but it kept 'V9', which has the p-value of 0.10. We test this model with 'drop1' function and see that 'V9' is not dropped based on the Chisquare test, so we will keep it in the model. All variables are indicated as significant.

c-) Conduct the Hosmer-Lemeshow goodness of fit test for the appropriateness of the logistic regression function by forming four groups. State the alternatives, decision rule, and conclusion.

```
hoslem.test(f2_step$y, fitted(f2_step), g=4)
```

```
##
##  Hosmer and Lemeshow goodness of fit (GOF) test
##
## data:  f2_step$y, fitted(f2_step)
## X-squared = 2.8127, df = 2, p-value = 0.245
```

H_0: The model is a good fit.
H_a: The model is not a good fit.

The p-value of the test above is 0.245, which is greater than the significance level, 0.05, so we accept the null hypothesis that the model is a good fit.

d-) What is the estimated probability of having a malignant tumor based on the data below? Copy and paste the code below on r.

```
test.dat<-data.frame(matrix(c(4 ,2,1,1,2,1,2,1,1,8,7,5,10,7,9,5,5,4,6 ,1,1,1,2,1,3,1,1),byrow=T,nrow=3,
dimnames(test.dat)[[2]]<-c("V1","V2","V3","V4","V5","V6","V7","V8","V9")

pred <- predict(f2_step, test.dat, type="link", se.fit = TRUE)
```

```
inv.logit(pred$fit)
```

```
##            1            2            3
## 0.9938770314 0.0002034273 0.9723373528
```

The estimated probabilities of having malignant tumor are shown above. Since we set 0 as malignant, the predicted value closer to 0 means a higher chance of having a malignant tumor. Closer to 1 means benign.

Question 3

Use dvisits data sets in the faraway package. Copy and paste and following code "library(faraway);data(dvisits,package ="faraway")". dvisits. The depdent variable is doctorco which is Number of consultations with a doctor or specialist in the past 2 weeks. You can"help(dvisits)" command in R to see data describtions of the reamining variables.

The response variable is doctorco and there are 8 independent variables. pollib is the only categorical variable in the data set.

(15 points, 5 points each)

a-) Fit a regression model with the doctorco variable as the response and all other variables except **nonpresc, and nondocco (use "dvisits[-c(14,19)]" to drop these variables in R)** are predictors (dummy variables are already created,use the data as it is). Is the model significant?

```
data(dvisits)
str(dvisits)
```

```
## 'data.frame':    5190 obs. of  19 variables:
##  $ sex     : int  1 1 0 0 0 1 1 1 1 0 ...
##  $ age     : num  0.19 0.19 0.19 0.19 0.19 0.19 0.19 0.19 0.19 0.19 ...
##  $ agesq   : num  0.0361 0.0361 0.0361 0.0361 0.0361 0.0361 0.0361 0.0361 0.0361 0.0361 ...
##  $ income  : num  0.55 0.45 0.9 0.15 0.45 0.35 0.55 0.15 0.65 0.15 ...
##  $ levyplus: int  1 1 0 0 0 0 0 0 1 1 ...
##  $ freepoor: int  0 0 0 0 0 0 0 0 0 0 ...
##  $ freerepa: int  0 0 0 0 0 0 0 0 0 0 ...
##  $ illness : int  1 1 3 1 2 5 4 3 2 1 ...
##  $ actdays : int  4 2 0 0 5 1 0 0 0 0 ...
##  $ hscore  : int  1 1 0 0 1 9 2 6 5 0 ...
##  $ chcond1 : int  0 0 0 0 1 1 0 0 0 0 ...
##  $ chcond2 : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ doctorco: int  1 1 1 1 1 1 1 1 1 1 ...
##  $ nondocco: int  0 0 0 0 0 0 0 0 0 0 ...
##  $ hospadmi: int  0 0 1 0 0 0 0 0 0 0 ...
##  $ hospdays: int  0 0 4 0 0 0 0 0 0 0 ...
##  $ medicine: int  1 2 2 0 3 1 0 1 1 1 ...
##  $ prescrib: int  1 1 1 0 1 1 0 1 0 1 ...
##  $ nonpresc: int  0 1 1 0 2 0 0 0 1 0 ...
```

```
head(dvisits)
```

```
##   sex  age  agesq income levyplus freepoor freerepa illness actdays hscore
## 1   1 0.19 0.0361   0.55        1        0        0       1       4      1
## 2   1 0.19 0.0361   0.45        1        0        0       1       2      1
## 3   0 0.19 0.0361   0.90        0        0        0       3       0      0
## 4   0 0.19 0.0361   0.15        0        0        0       1       0      0
## 5   0 0.19 0.0361   0.45        0        0        0       2       5      1
## 6   1 0.19 0.0361   0.35        0        0        0       5       1      9
```

```
##   chcond1 chcond2 doctorco nondocco hospadmi hospdays medicine prescrib
## 1       0       0        1        0        0        0        1        1
## 2       0       0        1        0        0        0        2        1
## 3       0       0        1        0        1        4        2        1
## 4       0       0        1        0        0        0        0        0
## 5       1       0        1        0        0        0        3        1
## 6       1       0        1        0        0        0        1        1
##   nonpresc
## 1        0
## 2        1
## 3        1
## 4        0
## 5        2
## 6        0
```

```
apply(dvisits,2,function(x) sum(is.na(x)))
```

```
##      sex      age    agesq   income levyplus freepoor freerepa  illness
##        0        0        0        0        0        0        0        0
##  actdays   hscore  chcond1  chcond2 doctorco nondocco hospadmi hospdays
##        0        0        0        0        0        0        0        0
## medicine prescrib nonpresc
##        0        0        0
```

```
dvisits2 <- dvisits[-c(14,19)]
f3 <- glm(doctorco ~ ., family=poisson, data=dvisits2)
summary(f3)
```

```
##
## Call:
## glm(formula = doctorco ~ ., family = poisson, data = dvisits2)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -4.0577  -0.6626  -0.5717  -0.5072   5.5949
##
## Coefficients:
##              Estimate Std. Error z value       Pr(>|z|)
## (Intercept) -2.076121   0.191496 -10.842        < 2e-16 ***
## sex          0.128448   0.056545   2.272         0.0231 *
## age          0.870301   1.014154   0.858         0.3908
## agesq       -1.060764   1.091992  -0.971         0.3313
## income      -0.131424   0.089699  -1.465         0.1429
## levyplus     0.055261   0.072430   0.763         0.4455
## freepoor    -0.430511   0.180107  -2.390         0.0168 *
## freerepa     0.021119   0.093021   0.227         0.8204
## illness      0.134461   0.019955   6.738 0.000000000016 ***
## actdays      0.112144   0.005379  20.849        < 2e-16 ***
## hscore       0.018155   0.010260   1.769         0.0768 .
## chcond1      0.078950   0.068083   1.160         0.2462
## chcond2     -0.022833   0.086867  -0.263         0.7927
## hospadmi     0.157641   0.031328   5.032 0.000000485280 ***
## hospdays     0.001297   0.002909   0.446         0.6557
## medicine    -0.092332   0.037582  -2.457         0.0140 *
## prescrib     0.228126   0.038775   5.883 0.000000004020 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 5634.8  on 5189  degrees of freedom
## Residual deviance: 4246.4  on 5173  degrees of freedom
## AIC: 6612
##
## Number of Fisher Scoring iterations: 6
```

There is no missing value. We fit the poisson regression model since the response variable is count data. 'sex','freepoor','illness','hospadmi','medicine','prescrib' are indicated as significant variables. The model is significant because the residual deviance of 4,246.4 is reduced by 1,388.4 from the null deviance of 5,634.8.

b-) Perform variable selection (Hint: Use the step function.) **Ensure that all variables are significant, use $\alpha = 0.05$.** Calculate the SSE for the model.

```r
f3_step <- step(f3, direction="both", trace=0)
summary(f3_step)
```

```
##
## Call:
## glm(formula = doctorco ~ sex + freepoor + illness + actdays +
##     hscore + chcond1 + hospadmi + medicine + prescrib, family = poisson,
##     data = dvisits2)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -4.1592  -0.6658  -0.5693  -0.5195   5.5865
##
## Coefficients:
##              Estimate Std. Error z value         Pr(>|z|)
## (Intercept) -2.002890   0.055273 -36.237          < 2e-16 ***
## sex          0.144502   0.054204   2.666          0.00768 **
## freepoor    -0.415783   0.172010  -2.417          0.01564 *
## illness      0.133793   0.019635   6.814 0.00000000000948 ***
## actdays      0.111725   0.005281  21.158          < 2e-16 ***
## hscore       0.019309   0.010159   1.901          0.05736 .
## chcond1      0.084180   0.053343   1.578          0.11454
## hospadmi     0.165526   0.029043   5.699 0.00000001202714 ***
## medicine    -0.090567   0.037249  -2.431          0.01504 *
## prescrib     0.226339   0.037593   6.021 0.00000000173606 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 5634.8  on 5189  degrees of freedom
## Residual deviance: 4249.7  on 5180  degrees of freedom
## AIC: 6601.3
##
## Number of Fisher Scoring iterations: 6
```

```r
drop1(f3_step, test="Chi")
```

```
## Single term deletions
##
## Model:
## doctorco ~ sex + freepoor + illness + actdays + hscore + chcond1 +
##     hospadmi + medicine + prescrib
##          Df Deviance    AIC    LRT         Pr(>Chi)
## <none>        4249.7 6601.3
## sex       1   4256.9 6606.5   7.17         0.007393 **
## freepoor  1   4256.4 6606.0   6.67         0.009809 **
## illness   1   4295.3 6644.9  45.59 0.00000000001456 ***
## actdays   1   4633.4 6983.0 383.70        < 2.2e-16 ***
## hscore    1   4253.3 6602.9   3.55         0.059641 .
## chcond1   1   4252.2 6601.8   2.49         0.114883
## hospadmi  1   4279.6 6629.2  29.88 0.00000004590963 ***
## medicine  1   4256.1 6605.6   6.31         0.011992 *
## prescrib  1   4291.9 6641.5  42.17 0.00000000008370 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
f3_best <- glm(doctorco ~ sex+freepoor+illness+actdays+hospadmi+medicine+prescrib,
               family=poisson, data=dvisits2)
summary(f3_best)
```

```
##
## Call:
## glm(formula = doctorco ~ sex + freepoor + illness + actdays +
##     hospadmi + medicine + prescrib, family = poisson, data = dvisits2)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -4.0780  -0.6571  -0.5698  -0.5281   5.5933
##
## Coefficients:
##              Estimate Std. Error z value           Pr(>|z|)
## (Intercept) -1.970180   0.052423 -37.582            < 2e-16 ***
## sex          0.151966   0.053949   2.817            0.00485 **
## freepoor    -0.406554   0.171759  -2.367            0.01793 *
## illness      0.145528   0.019011   7.655 0.0000000000000193 ***
## actdays      0.113875   0.005025  22.663            < 2e-16 ***
## hospadmi     0.165500   0.028303   5.847 0.0000000049916774 ***
## medicine    -0.087906   0.037162  -2.365            0.01801 *
## prescrib     0.227641   0.037533   6.065 0.0000000013190532 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 5634.8  on 5189  degrees of freedom
## Residual deviance: 4255.3  on 5182  degrees of freedom
## AIC: 6602.9
##
## Number of Fisher Scoring iterations: 6
```

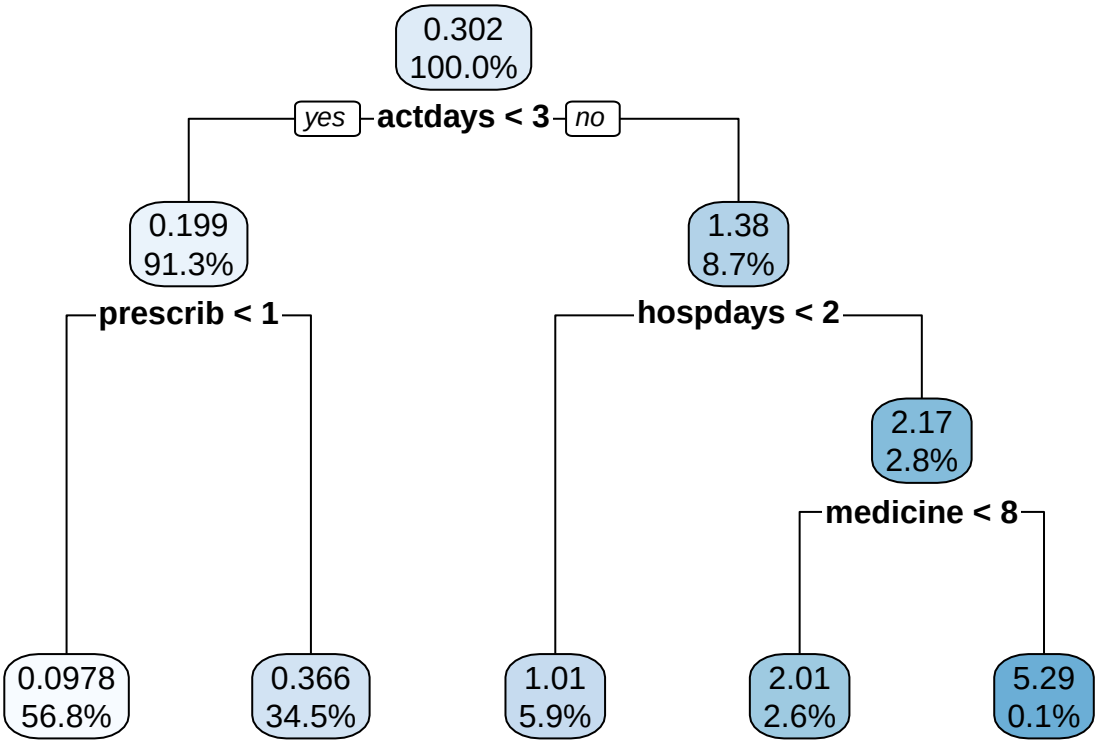
```
SSE(dvisits2$doctorco, predict(f3_best, dvisits2))
```

```
## [1] 19383.82
```

After we use the step function for the variable selection, we still see that 'hscore' and 'chcond1' as insignificant variables with p-value over 0.05. Even after using drop1 function, those two variables are indicated as insignificant, so we manually dropped them. The final model has all variables as significant, and the SSE = 19,383.82.

c-) Use regression tree approach to predict the doctorco and calculate SSE and compare it against SSE in part b. which model would you choose?

```
f3tree <- rpart(doctorco ~ ., data=dvisits2)
rpart.plot(f3tree, digits=3)
```




```
SSE(dvisits2$doctorco, predict(f3tree, dvisits2))
```

```
## [1] 2450.206
```

The SSE for the regression tree is 2,450.206, which is smaller than the SSE from part(b), so we would choose the regression tree model.